

INTERNATIONAL A-LEVEL FURTHER MATHEMATICS FM03

(9665/FM03) Unit FP2 Pure Mathematics

Mark scheme

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2 5 1 X F M 0 3 / M S

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Key to mark scheme abbreviations

M	Mark is for method
m	Mark is dependent on one or more M marks and is for method
A	Mark is dependent on M or m marks and is for accuracy
B	Mark is independent of M or m marks and is for method and accuracy
E	Mark is for explanation
✓ or ft	Follow through from previous incorrect result
CAO	Correct answer only
CSO	Correct solution only
AWFW	Anything which falls within
AWRT	Anything which rounds to
ACF	Any correct form
AG	Answer given
SC	Special case
oe	Or equivalent
A2, 1	2 or 1 (or 0) accuracy marks
–x EE	Deduct x marks for each error
NMS	No method shown
PI	Possibly implied
SCA	Substantially correct approach
sf	Significant figure(s)
dp	Decimal place(s)
ISW	Ignore subsequent working

Q	Answer	Marks	Comments
1	$\overrightarrow{AB} = \begin{bmatrix} k+1 \\ 2 \\ 1 \end{bmatrix}, \quad \overrightarrow{AC} = \begin{bmatrix} 3 \\ 3 \\ 1 \end{bmatrix}, \quad \overrightarrow{AD} = \begin{bmatrix} k+2 \\ -1 \\ 3 \end{bmatrix}$ $\overrightarrow{AB} \times \overrightarrow{AC} \bullet \overrightarrow{AD} = \begin{bmatrix} k+1 \\ 2 \\ 1 \end{bmatrix} \times \begin{bmatrix} 3 \\ 3 \\ 1 \end{bmatrix} \bullet \begin{bmatrix} k+2 \\ -1 \\ 3 \end{bmatrix}$ $\overrightarrow{AB} \times \overrightarrow{AC} \bullet \overrightarrow{AD} = \begin{bmatrix} -1 \\ -k+2 \\ 3k-3 \end{bmatrix} \bullet \begin{bmatrix} k+2 \\ -1 \\ 3 \end{bmatrix}$ $\overrightarrow{AB} \times \overrightarrow{AC} \bullet \overrightarrow{AD} = 9k - 13$ $9k - 13 = 4$ $k = \frac{17}{9}$	B1 M1 A1 A1 M1 A1	Obtains three non-coplanar direction vectors, eg $\overrightarrow{AB}$, $\overrightarrow{AC}$ and $\overrightarrow{AD}$ Calculates scalar triple product of their three direction vectors. Do not accept position vectors Correct vector product Correct scalar triple product. Do not accept in vector form Their scalar triple product = $(\pm)4$ Correct values of k
	Question 1 Total	6	

Q	Answer	Marks	Comments
2(a)	$\cosh^2 x - \sinh^2 x = \left\{ \frac{1}{2} (e^x + e^{-x}) \right\}^2 - \left\{ \frac{1}{2} (e^x - e^{-x}) \right\}^2$ $= \frac{1}{4} (e^{2x} + 2 + e^{-2x}) - \frac{1}{4} (e^{2x} - 2 + e^{-2x})$ $= 1$	M1 A1	Uses correct expressions for $\cosh x$ and $\sinh x$ to simplify $\cosh^2 x - \sinh^2 x$ Correct working to required answer AG Must include the LHS, at least two intermediate steps, and the RHS
		2	

Q	Answer	Marks	Comments
2(b)	$\cosh^2 x - 1 - \cosh x - 5 = 0$ $\cosh^2 x - \cosh x - 6 = 0$ $\cosh x = 3$ or $\cosh x = -2$ $\cosh x \neq -2$ [as $\cosh x \geq 1$] $x = \pm \cosh^{-1}(3)$ $x = \ln(3 \pm \sqrt{8})$	M1 M1 A1 E1 A1	Correct use of identity from part (a) Forms three term quadratic equation in $\cosh x$ PI Correct solution(s) Correct answers from correct quadratic. Must see both. Rejection of $\cosh x = -2$ Both correct solutions and no others, ACF Eg $x = \pm \ln(3 + \sqrt{8})$ Must score M1M1A1 to get final A1 Can score B1M1A1E0A1
		5	

	Question 2 Total	7	
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Q	Answer	Marks	Comments
3(a)	$\mathbf{R}^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & \sin\theta \\ 0 & -\sin\theta & \cos\theta \end{bmatrix}$ $\mathbf{A} = \mathbf{B}\mathbf{R}^{-1}$ $\mathbf{A} = \begin{bmatrix} 0 & \cos\theta & -\sin\theta \\ 1 & 0 & 0 \\ 0 & \sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & \sin\theta \\ 0 & -\sin\theta & \cos\theta \end{bmatrix}$ $\mathbf{A} = \begin{bmatrix} 0 & \cos^2\theta + \sin^2\theta & 0 \\ 1 & 0 & 0 \\ 0 & 0 & \cos^2\theta + \sin^2\theta \end{bmatrix}$ $\mathbf{A} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ $\mathbf{A}$ is independent of θ	B1 M1 A1	Obtains correct matrix for $\mathbf{R}^{-1}$ Correct method to find $\mathbf{A}$, by right multiplication by $\mathbf{R}^{-1}$, can be algebraic Correct result including conclusion that $\mathbf{A}$ is independent of θ
		3	

Q	Answer	Marks	Comments
3(b)	Reflection in $y = x$ plane	M1 A1	Reflection Correct plane SC1 for 'reflect' or 'reflected' or 'reflects' in $y = x$ plane
		2	
Question 3 Total		5	

Q	Answer	Marks	Comments
4(a)	The integrand $\frac{1}{\sqrt{4-x^2}}$ is not defined at $x = 2$	E1	oe
		1	

Q	Answer	Marks	Comments
4(b)	$\int \frac{1}{\sqrt{4-x^2}} dx = \sin^{-1}\left(\frac{x}{2}\right)$ $\int_0^2 \frac{1}{\sqrt{4-x^2}} dx = \lim_{a \rightarrow 2} \int_0^a \frac{1}{\sqrt{4-x^2}} dx$ $= \lim_{a \rightarrow 2} \left[\sin^{-1}\left(\frac{x}{2}\right) \right]_0^a$ $\int_0^2 \frac{1}{\sqrt{4-x^2}} dx = \lim_{a \rightarrow 2} \left[\sin^{-1}\left(\frac{a}{2}\right) - \sin^{-1}(0) \right]$ $= \sin^{-1}(1)$ $= \frac{\pi}{2}$	B1 M1 A1	Correct integration, with or without limits Evidence of limit 2 replaced by a and limit taken somewhere oe $\lim_{a \rightarrow 2}$ seen at any stage Do not accept the limit as x Correct with limiting process clearly shown. Do not accept 90°
		3	

	Question 4 Total	4	
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Q	Answer	Marks	Comments
5	$f(1) = 7^2 + 11^1 = 60 [= 4 \times 15]$ $[\text{True for } n = 1]$ $n = k + 1$ Assume true $n = k$ (*) Makes 11^k or 7^{k+1} the subject of an equation eg $11^k = 4M - 7^{k+1}$ (I) or Considers $f(k+1) + p f(k) = \dots$ $f(k+1) = 7^{k+2} + 11^{k+1}$ $= 7^{k+2} + 11(4M - 7^{k+1})$ or Rearranges to get $f(k+1) = \dots - p f(k)$ $f(k+1) =$ a correct function that is a multiple of 4 Hence true for $n = k+1$ (**) and since true for $n = 1$ (***) By induction $f(n)$ is a multiple of 4 for all integers $n \geq 1$ (****)	B1 M1 M1 A1 E1	$f(k) = 7^{k+1} + 11^k = 4M$ $7^{k+1} = 4M - 11^k$ (I) Do not condone $q f(k+1) + p f(k) \quad q \neq 1$ Shows that $f(k+1)$ is a multiple of 4 $f(k+1) = 4 \times 11M + 7^{k+1}(7 - 11)$ Eg $= 4(11M - 7^{k+1})$ Must include $f(k+1) =$ Must have (*), (**), (***) present, previous 4 marks scored and a final statement (****) clearly indicating that it relates to all integers $n \geq 1$ and induction
	Question 5 Total	5	

Q	Answer	Marks	Comments
6(a)	$(r+1)^5 - (r-1)^5$ $= r^5 + 5r^4 + 10r^3 + 10r^2 + 5r + 1$ $- (r^5 - 5r^4 + 10r^3 - 10r^2 + 5r - 1)$ $= 10r^4 + 20r^2 + 2$	B1	Correct result from correct working, AG . Must have LHS, RHS and at least one line of intermediate working
		1	

Q	Answer	Marks	Comments
6(b)	$S_n = \sum_{r=1}^n \left\{ (r+1)^5 - (r-1)^5 \right\}$ $= \cancel{2^5} - 0^5$ $+ \cancel{3^5} - 1^5$ $+ \cancel{4^5} - \cancel{2^5}$ $\vdots$ $+ \cancel{(n-1)^5} - \cancel{(n-3)^5}$ $+ n^5 - \cancel{(n-2)^5}$ $+ (n+1)^5 - \cancel{(n-1)^5}$ $= n^5 + (n+1)^5 - 1$ States or uses $\sum_{r=1}^n r^2 = \frac{1}{6}n(n+1)(2n+1)$ and $\sum_{r=1}^n 1 = n$ $10 \sum_{r=1}^n r^4 = n^5 + (n+1)^5 - 1$ $- 20 \times \frac{1}{6}n(n+1)(2n+1) - 2n$ $10 \sum_{r=1}^n r^4 = 2n^5 + 5n^4 + 10n^3 + 10n^2 + 3n$ $- 20 \times \frac{1}{6}n(n+1)(2n+1)$ $10 \sum_{r=1}^n r^4 = n(n+1)(2n^3 + 3n^2 + 7n + 3)$ $- \frac{10}{3}n(n+1)(2n+1)$ $\sum_{r=1}^n r^4 = \frac{1}{30}n(n+1)(2n+1)(3n^2 + 3n - 1)$	M1 A1 B1 m1 m1 m1 A1	Uses method of differences showing the first and last lines and at least two other lines so that a pair of values which cancel are seen Correct answer Use of $\sum_{r=1}^n r^2$, $\sum_{r=1}^n 1$ and their S_n to obtain an equation for $\sum_{r=1}^n r^4$ $\sum_{r=1}^n 1$ and their S_n simplified Attempts to factorise out $n(n+1)$ or $n(2n+1)$ or $(n+1)(2n+1)$ Correct result from correct working
		7	
	Question 6 Total	8	

Q	Answer	Marks	Comments
7	$\text{Area} = \frac{1}{2} \int_0^{\frac{\pi}{6}} 2(1 + \tan \theta) \sec^4 \theta \, d\theta$ $= \int_0^{\frac{\pi}{6}} \sec^4 \theta \, d\theta + \int_0^{\frac{\pi}{6}} \tan \theta \sec^4 \theta \, d\theta$ $\int_0^{\frac{\pi}{6}} \sec^4 \theta \, d\theta = \int_0^{\frac{\pi}{6}} (1 + \tan^2 \theta) \sec^2 \theta \, d\theta$ $= \int_0^{\frac{\pi}{6}} (\sec^2 \theta + \tan^2 \theta \sec^2 \theta) \, d\theta$ $\int_0^{\frac{\pi}{6}} \tan \theta \sec^4 \theta \, d\theta = \int_0^{\frac{\pi}{6}} (\tan \theta \sec \theta) \sec^3 \theta \, d\theta$ $\text{Area} = \left[\tan \theta + \frac{1}{3} \tan^3 \theta + \frac{1}{4} \sec^4 \theta \right]_0^{\frac{\pi}{6}}$ $\text{Area} = \left[\frac{\sqrt{3}}{3} + \frac{1}{3} \left(\frac{\sqrt{3}}{3} \right)^3 + \frac{1}{4} \left(\frac{2\sqrt{3}}{3} \right)^4 - \frac{1}{4} \right]$ $= \frac{21 + 40\sqrt{3}}{108}$	B1 M1 M1 A1 A1	$\frac{1}{2} \int_0^{\frac{\pi}{6}} r^2 \, d\theta$ used with correct integrand, correct limits and $d\theta$. All seen here Valid method to integrate $\sec^4 \theta$ Valid method to integrate $\tan \theta \sec^4 \theta$ Integrates to obtain $p \tan \theta + q \tan^3 \theta + r \sec^4 \theta$ Can score B0 M1M1A1A1

Q	Answer	Marks	Comments
7 ALT	$\text{Area} = \frac{1}{2} \int_0^{\frac{\pi}{6}} 2(1 + \tan \theta) \sec^4 \theta \, d\theta$ $t = \tan \theta$ $dt = \sec^2 \theta \, d\theta$ $\text{Area} = \int_0^{\frac{1}{\sqrt{3}}} (1+t)(1+t^2) \, dt$ $\text{Area} = \int_0^{\frac{1}{\sqrt{3}}} (1+t+t^2+t^3) \, dt$ $= \left[t + \frac{1}{2}t^2 + \frac{1}{3}t^3 + \frac{1}{4}t^4 \right]_0^{\frac{1}{\sqrt{3}}}$ $\text{Area} = \left[\frac{1}{\sqrt{3}} + \frac{1}{2} \left(\frac{1}{\sqrt{3}} \right)^2 + \frac{1}{3} \left(\frac{1}{\sqrt{3}} \right)^3 + \frac{1}{4} \left(\frac{1}{\sqrt{3}} \right)^4 \right]$ $= \frac{21 + 40\sqrt{3}}{108}$	B1 M1 A1 A1 A1	$\frac{1}{2} \int_0^{\frac{\pi}{6}} r^2 \, d\theta$ used with correct integrand, correct limits and $d\theta$. All seen here Valid method using $t = \tan \theta$ to integrate $k \int_0^{\frac{\pi}{6}} (1 + \tan \theta) \sec^4 \theta \, d\theta$. Condone lack of or wrong limits Correct Integration. Condone lack of or wrong limits Correct limits, seen at any point PI correct substitution Can score B0 M1A1A1A1
	Question 7 Total	5	

Q	Answer	Marks	Comments
8(a)	$0 = \begin{vmatrix} 2 & -1 & 3 \\ 1 & k-1 & -1 \\ k-3 & -1 & 1 \end{vmatrix}$ $= 2(k-1-1) + (1+k-3) + 3\{-1-(k-1)(k-3)\}$ $0 = 3k - 6 - 12 + 12k - 3k^2$ $= -3k^2 + 15k - 18$ $0 = k^2 - 5k + 6$ $k = 2 \text{ or } k = 3$	M1 B1 A1	Sets determinant = 0 and attempts to expand Obtains a correct quadratic expression Obtains correct values of k
		3	

Q	Answer	Marks	Comments
8(b)(i)	$\begin{array}{rrcr} 2x & -y & +3z & = & 1 & \text{(I)} \\ x & +2y & -z & = & 3 & \text{(II)} \\ & -y & +z & = & 1 & \text{(III)} \end{array}$ $\begin{array}{l} \text{(I)} - 2\text{(II)} \Rightarrow -5y + 5z = -5 \\ 5\text{(III)} \Rightarrow -5y + 5z = 5 \end{array}$ (Inconsistent) no solutions	B1 M1 A1	Correct system of equations in the case $k = 3$ Eliminating one variable in order to compare two simultaneous equations From comparing correct equations
		3	

Q	Answer	Marks	Comments
8(b)(ii)	Three planes form a (triangular) prism	E1	Must come from correct working in part (b)(i)
		1	

	Question 8 Total	7	
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Q	Answer	Marks	Comments
9	$\frac{dx}{d\theta} = \cos^2\theta - \sin^2\theta$ $\frac{dy}{d\theta} = 2\sin\theta\cos\theta$ $\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2 = \cos^4\theta - 2\sin^2\theta\cos^2\theta + \sin^4\theta + 4\sin^2\theta\cos^2\theta$ $\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2 = \cos^4\theta + 2\sin^2\theta\cos^2\theta + \sin^4\theta = (\cos^2\theta + \sin^2\theta)^2 = 1$ $S = 2\pi \int_0^{\frac{\pi}{6}} \sin^2\theta \times 1 d\theta$ $= \pi \int_0^{\frac{\pi}{6}} (1 - \cos 2\theta) d\theta$ $= \pi \left[\theta - \frac{1}{2} \sin 2\theta \right]_0^{\frac{\pi}{6}}$ $= \pi \left[\frac{\pi}{6} - \frac{1}{2} \times \frac{\sqrt{3}}{2} \right]$ $= \frac{\pi}{12} [2\pi - 3\sqrt{3}]$	B1 M1 A1 B1ft M1 M1 M1 A1	Correct expression for $\frac{dx}{d\theta}$ and $\frac{dy}{d\theta}$ Correctly expands their $\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2$ $\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2 = 1$ from correct working Their correct integrand, with 2π, $d\theta$ and correct limits. All seen here Use of $2\sin^2\theta = 1 - \cos 2\theta$ Correctly integrating $k(1 \pm \cos 2\theta)$ Correct use of limits in an expression of the form $k(\theta \pm \frac{1}{2} \sin 2\theta)$ CSO – form of answer given
	Question 9 Total	8	

Q	Answer	Marks	Comments
10(a)	$(\beta - d) + \beta + (\beta + d) + (\beta + 2d) = \frac{2}{1}$ $4\beta + 2d = 2$ $\frac{-21}{1} = (\beta - d)\beta + (\beta - d)(\beta + d)$ $+ (\beta - d)(\beta + 2d) + \beta(\beta + d)$ $+ \beta(\beta + 2d) + (\beta + d)(\beta + 2d)$ $-21 = 6\beta^2 + 6\beta d - d^2$ $10\beta^2 - 10\beta - 20 = 0$ $\beta = 2, d = -3 \text{ or } \beta = -1, d = 3$ $\alpha = 5 \qquad \alpha = -4$ $\beta = 2 \qquad \text{or} \qquad \beta = -1$ $\gamma = -1 \qquad \gamma = 2$ $\delta = -4 \qquad \delta = 5$	M1 Use of $\sum a = \frac{-b}{a}$ with arithmetic sequence structure. If $\alpha, \alpha + d, \alpha + 2d, \alpha + 3d$ $4\alpha + 6d = 2$ Condone -2 instead of 2 M1 Use of $\sum a\beta = \frac{c}{a}$ with arithmetic sequence structure. If $\alpha, \alpha + d, \alpha + 2d, \alpha + 3d$ $-21 = 6\alpha^2 + 18\alpha d + 11d^2$ Condone $+21$ instead of -21 A1 Obtains two correct, simultaneous equations M1 Eliminates one variable to get a three term quadratic equation If $\alpha, \alpha + d, \alpha + 2d, \alpha + 3d$ $10\alpha^2 - 10\alpha - 200 = 0$ A1 Finds correct β and d If $\alpha, \alpha + d, \alpha + 2d, \alpha + 3d$ $\alpha = 5, d = -3$ or $\alpha = -4, d = 3$ A1 Correct values of α, β, γ and δ	Use of $\sum a = \frac{-b}{a}$ with arithmetic sequence structure. If $\alpha, \alpha + d, \alpha + 2d, \alpha + 3d$ $4\alpha + 6d = 2$ Condone -2 instead of 2 Use of $\sum a\beta = \frac{c}{a}$ with arithmetic sequence structure. If $\alpha, \alpha + d, \alpha + 2d, \alpha + 3d$ $-21 = 6\alpha^2 + 18\alpha d + 11d^2$ Condone $+21$ instead of -21 Obtains two correct, simultaneous equations Eliminates one variable to get a three term quadratic equation If $\alpha, \alpha + d, \alpha + 2d, \alpha + 3d$ $10\alpha^2 - 10\alpha - 200 = 0$ Finds correct β and d If $\alpha, \alpha + d, \alpha + 2d, \alpha + 3d$ $\alpha = 5, d = -3$ or $\alpha = -4, d = 3$ Correct values of α, β, γ and δ
		6	

Q	Answer	Marks	Comments
10(b)	$\frac{-p}{1} = 5 \times 2 \times -1 + 5 \times 2 \times -4$ $+ 5 \times -1 \times -4 + 2 \times -1 \times -4$ $p = 22$	M1	Use of $\sum \alpha\beta\gamma = \frac{-d}{a}$ and their roots to get a value of p Condone p instead of $-p$
	$\frac{q}{1} = 5 \times 2 \times -1 \times -4$ $q = 40$	M1	Use of $\alpha\beta\gamma\delta = \frac{e}{a}$ and their roots to get a value of q Condone $-q$ instead of q
	$p = 22$ $q = 40$	A1	Obtains the correct values of p and q If 0/3 scored SC1 for writing $p = -\sum \alpha\beta\gamma$ and $q = \alpha\beta\gamma\delta$
		3	
	Question 10 Total	9	

Q	Answer	Marks	Comments
11(a)	$\mathbf{r}_1 = \begin{bmatrix} 6 \\ 7 \\ 5 \end{bmatrix} + \lambda \begin{bmatrix} 0.5 \\ 2 \\ -1 \end{bmatrix}$ $\mathbf{r}_2 = \begin{bmatrix} 8 \\ w \\ 6 \end{bmatrix} + \mu \begin{bmatrix} -3 \\ 2 \\ 1 \end{bmatrix}$ $\begin{bmatrix} 6 \\ 7 \\ 5 \end{bmatrix} + \lambda \begin{bmatrix} 0.5 \\ 2 \\ -1 \end{bmatrix} = \begin{bmatrix} 8 \\ w \\ 6 \end{bmatrix} + \mu \begin{bmatrix} -3 \\ 2 \\ 1 \end{bmatrix}$ x $0.5\lambda + 3\mu = 2$ z $\lambda + \mu = -1$ $\lambda = -2, \mu = 1$ y $7 - 2 \times 2 = w + 1 \times 2$ $w = 1$	B1 B1 M1 A1 A1	Correct general point for L_1 Correct general point for L_2 Condone same parametric variable as used for L_1 Uses x and z general points to form a pair of simultaneous equations in two different parametric variables Correct values for the parametric variables Correct w
		5	

Q	Answer	Marks	Comments
11(b)	$\mathbf{n} = \begin{bmatrix} 0.5 \\ 2 \\ -1 \end{bmatrix} \times \begin{bmatrix} -3 \\ 2 \\ 1 \end{bmatrix}$ $d = \begin{bmatrix} 8 \\ 5 \\ 14 \end{bmatrix} \cdot \begin{bmatrix} 6 \\ 7 \\ 5 \end{bmatrix} = 153$ $\mathbf{r} \cdot \begin{bmatrix} 8 \\ 5 \\ 14 \end{bmatrix} = 153$	M1 M1 A1	Vector product of their direction vectors from part (a) Forms the scalar product of multiple of their normal vector and a position vector of a point in Π Condone use of a point on either of their lines A correct equation of Π, in the specified form.
		3	

Q	Answer	Marks	Comments
11(c)(i)	$153 = \begin{bmatrix} 5+8\sigma \\ -3+5\sigma \\ 1+14\sigma \end{bmatrix} \cdot \begin{bmatrix} 8 \\ 5 \\ 14 \end{bmatrix}$ $= 40 + 64\sigma - 15 + 25\sigma + 14 + 196\sigma$ $\sigma = \frac{2}{5}$ $\sigma' = \frac{4}{5}$ $Q = \begin{bmatrix} \frac{57}{5} \\ 5 \\ 1 \\ \frac{61}{5} \\ 5 \end{bmatrix}$	M1 A1 M1 A1	Substitutes equation of perpendicular line from P to Π into equation of Π Correct parametric variable PI $\left(\frac{41}{5}, -1, \frac{33}{5}\right)$ Doubles their parametric variable PI $\overline{PQ} = 2 \overline{PM}$ Correct position vector of Q Condone coordinates
		4	

Q	Answer	Marks	Comments
11(c)(ii)	$\frac{2}{5} \begin{bmatrix} 8 \\ 5 \\ 14 \end{bmatrix}$ $\frac{2}{5} \sqrt{285}$	M1 A1	Uses a correct method to find the required distance. Eg their $\sigma \times \mathbf{n}$ CAO
		2	

	Question 11 Total	14	
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Q	Answer	Marks	Comments
12	Auxiliary equation $m^2 - m - 2 = 0 \quad m = -1, 2$ $[y_{CF} =] \quad Ae^{-x} + Be^{2x}$ Particular Integral $[y_{PI} =] \quad C\cos x + D\sin x + Exe^{-x}$ $[y'_{PI} =] \quad -C\sin x + D\cos x + Ee^{-x} - Exe^{-x}$ $[y''_{PI} =] \quad -C\cos x - D\sin x - 2Ee^{-x} + Exe^{-x}$ $(-C\cos x - D\sin x - 2Ee^{-x} + Exe^{-x})$ $-(-C\sin x + D\cos x + Ee^{-x} - Exe^{-x})$ $-2(C\cos x + D\sin x + Exe^{-x})$ $= 6e^{-x} - 10\cos x$ $\cos x \qquad \sin x$ $3C + D = 10 \qquad C - 3D = 0$ $C = 3, D = 1$ e^{-x} $E = -2$ General Solution $[f(x) =] Ae^{-x} + Be^{2x} + 3\cos x + \sin x$ $\qquad \qquad \qquad - 2xe^{-x}$ $[f'(x) =] -Ae^{-x} + 2Be^{2x} - 3\sin x + \cos x$ $\qquad \qquad \qquad - 2e^{-x} + 2xe^{-x}$ $A + B = 7$ $-A + 2B = 2$ $A = 4, B = 3$ $f(x) = 4e^{-x} + 3e^{2x} + 3\cos x + \sin x - 2xe^{-x}$	M1 A1 B1 M1 M1 M1 M1 A1 B1 M1 M1 A1 A1	Finds solutions to the correct auxiliary equation Correct CF Correct particular integral Obtains first and second derivatives of their y_{PI} Substitutes their first and second derivatives of y_{PI} into DE and attempts to compare their coefficients Forms a pair of simultaneous equations from their coefficients of $\cos x$ and $\sin x$ Correct coefficients of $\cos x$ and $\sin x$ from correct equations Correct coefficient of xe^{-x} Differentiates their $f(x)$, but must have exactly two arbitrary constants Uses given conditions to form a pair of simultaneous equations for their two arbitrary constants Correct values for A and B, from correct equations Correct ISW
	Question 12 Total	12	

Q	Answer	Marks	Comments
13(a)(i)	$ M = 4(3 - k - 2) - (1 - 2k - 4) - (k + 1)(1 - 6)$ $= 12 + 3k$	M1	Obtains a linear expression for $ M $
	$k \neq -4$	A1	Correct restriction on k
		2	

Q	Answer	Marks	Comments
13(a)(ii)	Cofactor matrix		
	$\begin{bmatrix} 1-k & 3+2k & -5 \\ -2-k & 6+2k & -2 \\ 5+4k & -9-5k & 11 \end{bmatrix}$	M1	At least three entries correct
		A1	At least six entries correct
		A1	All nine entries correct
	Inverse matrix M^{-1}		
	$M^{-1} = \frac{1}{3(4+k)} \begin{bmatrix} 1-k & -2-k & 5+4k \\ 3+2k & 6+2k & -9-5k \\ -5 & -2 & 11 \end{bmatrix}$	M1	Transpose of their cofactors with no more than one further slip and division by their $ M $ in terms of k
		A1	Correct M^{-1}
		5	

Q	Answer	Marks	Comments
13(b)	$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{3(4+k)} \begin{bmatrix} 1-k & -2-k & 5+4k \\ 3+2k & 6+2k & -9-5k \\ -5 & -2 & 11 \end{bmatrix} \begin{bmatrix} 4 \\ 3 \\ 1 \end{bmatrix}$ $\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \frac{1}{3(4+k)} \begin{bmatrix} 4-4k-6-3k+5+4k \\ 12+8k+18+6k-9-5k \\ -20-6+11 \end{bmatrix}$ $x = \frac{1-k}{4+k}, \quad y = \frac{7+3k}{4+k}, \quad z = -\frac{5}{4+k}$	M1 A1ft A1	Multiplies their $\mathbf{M}^{-1}$ onto $\begin{bmatrix} 4 \\ 3 \\ 1 \end{bmatrix}$ to form a single unsimplified vector At least 2 entries correct for their $\mathbf{M}^{-1}$ Correct and fully simplified
		3	
	Question 13 Total	10	

Q	Answer	Marks	Comments
14(a)	$z - z^{-1} = 2i\sin\theta$	B1	Stated or used
	$(z - z^{-1})^6 = z^6 - 6z^4 + 15z^2 - 20 + 15z^{-2} - 6z^{-4} + z^{-6}$	M1	Expands $(z - z^{-1})^6$
	$(2i\sin\theta)^6 = z^6 + z^{-6} - 6(z^4 + z^{-4}) + 15(z^2 + z^{-2}) - 20$	A1	Forms a correct equation for $\sin^6\theta$ and expanded $(z - z^{-1})^6$
	$z^n + z^{-n} = 2\cos n\theta$	B1	Stated or used for $n = 2, 4$ or 6
	$-64\sin^6\theta = 2\cos 6\theta - 6 \times 2\cos 4\theta + 15 \times 2\cos 2\theta - 20$	M1	RHS of previous line collected and $z^n + z^{-n} = 2\cos n\theta$ used for $n = 2, 4$ and 6
	$\sin^6\theta = \frac{1}{32}(-\cos 6\theta + 6\cos 4\theta - 15\cos 2\theta + 10)$	A1	Correct answer from correct working
		6	

Q	Answer	Marks	Comments
14(b)	$\int_0^{\frac{\pi}{6}} \sin^6\theta \, d\theta = \frac{1}{32} \left[-\frac{1}{6}\sin 6\theta + \frac{6}{4}\sin 4\theta - \frac{15}{2}\sin 2\theta + 10\theta \right]_0^{\frac{\pi}{6}}$	M1	Integrates their answer to part (a) correctly, provided all terms in integrand are of the form $\cos(n\theta)$
	$= \frac{1}{32} \left(-\frac{1}{6} \times 0 + \frac{3}{2} \times \frac{\sqrt{3}}{2} - \frac{15}{2} \times \frac{\sqrt{3}}{2} + \frac{10\pi}{6} \right)$	A1	Correct integration
	$= \frac{1}{32} \left(-3\sqrt{3} + \frac{5\pi}{3} \right)$ $= \frac{1}{96} (5\pi - 9\sqrt{3})$	A1	Correct answer in the requested form, convincingly done AG
		3	

	Question 14 Total	9	
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Q	Answer	Marks	Comments
15(a)	$\frac{dy}{dx} + y \tan x = \frac{2x+5}{x^2+4x+5} \cos x$ $\text{I.F. is } e^{\int \tan x \, dx} = e^{\ln(\sec x)}$ $= \sec x$ $y \sec x = \int \left(\frac{2x+5}{x^2+4x+5} \right) dx$ $\frac{2x+5}{x^2+4x+5} = \frac{2x+4}{x^2+4x+5} + \frac{1}{x^2+4x+5}$ $y \sec x = \int \left(\frac{2x+4}{x^2+4x+5} \right) dx + \int \left(\frac{1}{(x+2)^2+1} \right) dx$ $\int \left(\frac{2x+4}{x^2+4x+5} \right) dx = \ln(x^2+4x+5)$ $\frac{1}{x^2+4x+5} = \frac{1}{(x+2)^2+1}$ $\int \left(\frac{1}{(x+2)^2+1} \right) dx = \tan^{-1}(x+2)$ $[f(x) =] \cos x \{ \ln(x^2+4x+5) + \tan^{-1}(x+2) + c \}$	M1 M1 A1 m1 M1 A1 M1 A1 A1	Dividing by $\cos x$ to obtain Standard Form Their I.F identified and integration attempted Correct integrating factor Multiplying both sides of the Standard Form by their I.F and integrating LHS to get $y \times \text{I.F}$ Splits RHS integrand into form $\frac{f'(x)}{f(x)} + \frac{A}{f(x)}$ Integrates $\frac{2x+4}{x^2+4x+5}$ to get $\ln(x^2+4x+5)$ Completes the square in the denominator of $\frac{A}{f(x)}$ Integrates $\frac{A}{(x+2)^2+1}$ to get $A \tan^{-1}(x+2)$ Condone lack of $+c$ Correct GS
		9	

Q	Answer	Marks	Comments
15(b)	$\tan^{-1}(2) = \ln(5) + \tan^{-1}(2) + c$ $c = -\ln(5)$	M1	Substituting $y = \tan^{-1}(2)$ and $x = 0$ into their GS (with a constant of integration) to find a value of c
	$y = \cos x \left\{ \ln(x^2 + 4x + 5) + \tan^{-1}(x + 2) - \ln(5) \right\}$ $= \cos x \left\{ \ln\left(\frac{x^2 + 4x + 5}{5}\right) + \tan^{-1}(x + 2) \right\}$	A1	Correct PS ACF , must be $y = \dots$
		2	
	Question 15 Total	11	